ALGEBRAIC FIBER SPACES WITH AMPLE RELATIVE ANTI-CANONICAL DIVISORS VIOLATING SUBADJUNCTION FORMULA ON MINIMAL LC CENTERS

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ABSTRACT. In this note, we construct algebraic fiber spaces $f : X \to Y$ of characteristic 2 or 3 from a projective 3-fold X to a smooth projective curve Y such that X has only canonical singularities (having crepant resolution) and $-K_{X/Y}$ is ample. Also, we construct algebraic fiber spaces $f : X \to Y$ such that $-K_{X/Y}$ is ample and X is lc whose unique lc center forms a section of f, which means that X violates the subadjunction formula.

1. Introduction

It is known that the relative anti-canonical divisor $-K_{X/Y}$ of an algebraic fiber space $f : X \to Y$ with mild singularities are not ample, unless the case when f is a constant map. At the first, Kollár–Miyaoka–Mori [6, Corollary 2.8] proved the case where f is a smooth morphism. Miyaoka [7, Theorem 2] showed the case when f is generically smooth. Araujo–Druel–Kovács [1, Theorem 3.1] extend these theorems to lc pairs (X, Δ) when Y is a curve.

From here on, we list up theorems that seems to be extended enough the above results. Let k be an algebraically closed field of characteristic $p \geq 0$. Let (X, Δ) be a pair and let $f : X \to Y$ be a separable surjective morphism to a smooth projective variety Y **of positive dimension**.

Theorem 1.1 ([3, Corollary 3.7])**.** *Assume that $p = 0$ and (X, Δ) is lc over the generic point of Y. Then, $-K_{X/Y} - \Delta$ is not ample.*

Corollary 1.2 ([2, Theorem 5.4])**.** *Assume that $p = 0$ and (X, Δ) is klt over the generic point of Y. Then $-K_{X/Y} - \Delta$ can not be both nef and big.*

Theorem 1.3 ([2, Corollary 4.10])**.** *Assume that $p > 0$ and $(X_{\overline{\eta}}, \Delta|_{X_{\overline{\eta}}})$ is F-pure, where $X_{\overline{\eta}}$ is the geometric generic fiber of f. Then $-K_{X/Y} - \Delta$ is not ample.*

Corollary 1.4 ([2, Corollary 4.10])**.** *Assume that $p > 0$ and $(X_{\overline{\eta}}, \Delta|_{X_{\overline{\eta}}})$ is strongly F-regular. Then $-K_{X/Y} - \Delta$ can not be both nef and big.*

Note that, when $p > 0$, we need to suppose that the geometric generic fiber has mild F-singularities. The F-purity (resp. strong F-regularity) of the geometric generic fiber implies the F-purity (resp. strong F-regularity) of generic fiber, but the inverse implication does not hold.

In this note, we consider the following question:

Question 1.5. Can we weaken the assumptions of Theorem 1.3 and Corollary 1.4 concerning the singularities of geometric generic fiber to the singularities of generic fiber?

We answer *negatively* this question:

Theorem 1.6. (1) (*Example 2.4, $p = 2, 3$*) *There exists an algebraic fiber space $f : X \to Y$ from a projective 3-fold X to a smooth projective curve Y such that X has only canonical singularities (having crepant resolution) and $-K_{X/Y}$ is ample.*
(2) (*Example 2.3, $p > 0$*) *There exists a strongly F-regular pair (P, Δ) of dimension 2 and an algebraic fiber space $f : P \to Y$ such that $-K_{P/Y} - \Delta$ is ample.*

Here, an algebraic fiber space is a separable surjective morphism with connected fibers. Although X in Theorem 1.6 is not F-pure, taking a crepant resolution $\tilde{X}$ of X, we can find an algebraic fiber space $\tilde{f} : \tilde{X} \to Y$ from a smooth projective 3-fold $\tilde{X}$ to a smooth projective curve Y such that $-K_{\tilde{X}/Y}$ is semi-ample (and nef) and big.

How to construct the algebraic fiber spaces in Theorem 1.6 is just adjustment the construction of Raynaud's surface ([9]) to the 3-dimensional case. A way slightly different from the above construction allows us to find lc 3-folds (without boundary) violating the subadjunction formula on minimal lc centers. Before stating the theorem, we recall the subadjunction formula:

Theorem 1.7 ([5, Theorem 1], [4, Theorem 1.2], $p = 0$)**.** *Let $\mathbb{K}$ be $\mathbb{Q}$ or $\mathbb{R}$. Let X be a normal projective variety and let D be an effective $\mathbb{K}$-divisor on X such that (X, D) is lc. Let W be a minimal lc center with respect to (X, D). Then*

$$(K_X + D)|_W \sim_{\mathbb{K}} K_W + D_W$$

for an effective $\mathbb{K}$-divisor D_W on W such that (W, D_W) is Kawamata log terminal. In particular, W has only rational singularities.

Theorem 1.8 (Example 2.5, $p = 2, 3$)**.** *Then there exists an algebraic fiber space $f : X \to Y$ from an lc 3-fold X to a smooth projective curve Y such that*

(1) *$-K_{X/Y}$ is ample and*
(2) *the unique lc center W forms a section of f.*

In particular, X violates the subadjunction formula on minimal lc centers, although W is smooth.

Indeed, we have

$$K_X|_W \sim (K_{X/Y} + f^*K_Y)|_W \sim \underbrace{K_{X/Y}|_W}_{\text{anti-ample}} + K_W,$$

so there exists no effective $\mathbb{K}$-divisor Δ_W with $K_X|_W \sim_{\mathbb{K}} K_W + \Delta_W$.

Declaration of AI assistance. The author used OpenAI's ChatGPT during the development of the example presented in this paper. In particular, ChatGPT was used to explore possible constructions and to assist with preliminary mathematical calculations. All mathematical statements and proofs presented in this paper have been independently verified by the author, who takes full responsibility for their correctness.

2. Algebraic fiber spaces with ample relative anti-canonical divisors

We first consider when Question 1.5 holds.

Notation 2.1. Let k be an algebraically closed field of characteristic $p > 0$. Let (X, Δ) be a projective pair and let $f : X \to Y$ be an algebraic fiber space to a smooth projective variety Y.

Lemma 2.2 ($p \geq 0$)**.** *Assume that* $\dim X - \dim Y = 1$ *and* $\mathrm{Supp}(\Delta)$ *does not dominate* Y*. Then* $-K_{X/Y} - \Delta$ *can not be both nef and big.*

Proof. Suppose that $-K_{X/Y}-\Delta$ is nef and big. Then $-K_{X_{\overline{\eta}}}$ is ample, so $X_{\overline{\eta}} \cong \mathbb{P}^1_{\overline{K(Y)}}$, and hence $(X_{\overline{\eta}}, \Delta_{X_{\overline{\eta}}})$ is klt (resp. strongly F-regular) when $p = 0$ (resp. $p > 0$). Note that $\Delta|_{X_{\overline{\eta}}} = 0$. Therefore, the statement follows from Theorems 1.2 and 1.4. □

By Lemma 2.2, when $\Delta = 0$, we see that there exists no algebraic fiber spaces such that $\dim X = 2$ and $-K_{X/Y}$ is both nef and big. Therefore, in the next step, we investigate in the following situation:

- The case when $\dim X = 2$ and $\Delta > 0$.
- The case when $\dim X = 3$, $\dim Y = 1$ and $\Delta = 0$.

Fortunately, in the both case, we can construct algebraic fiber space with mild singularities such that $-K_{X/Y} - \Delta$ is ample.

From here on, we use Notation 2.1. We use Tango–Raynaud curve $C = Y$ as base spaces. Let e, D and $C = Y$ be as in [8, Example 1.3]. Take $\xi \in H^1(Y, D) \setminus \{0\}$ so that $F^*\xi = 0$. Let

$$0 \to \mathcal{O}_Y(-D) \to \mathcal{E} \to \mathcal{O}_Y \to 0$$

be the extension corresponding to ξ. Then, by the choice of ξ, we find a subbundle $\mathcal{O}_Y \subset F^*\mathcal{E}$.

Example 2.3 ($p > 0$)**.** We construct a klt pair (P, Δ) of dimension 2 and a surjective morphism $f : P \to Y$ such that $-K_{P/Y}-\Delta$ is ample. Set $P := \mathbb{P}(\mathcal{E})$. Let $f : P \to Y$ be the projection. Put $T := c_1(\mathcal{O}_P(1))$. Let G' be the prime divisor on $\mathbb{P}(F^*\mathcal{E})$ corresponding to the subbundle $\mathcal{O}_Y \subset \mathcal{E}$. Set $G := F^*_{P/Y}G'$, where $F_{P/Y}$ is the relative Frobenius morphism of f. Then, by [8, PROPOSITION 1.7], we see that $\mathrm{Supp}(G)$ is smooth.

One can easily check that $G \sim pT$. Take an $\varepsilon \in \mathbb{Q}$ with $\frac{1}{p} < \varepsilon < \frac{2}{p} \leq 1$. Then $(P, \varepsilon G)$ is klt. Furthermore,

$$\begin{aligned} -K_{P/Y} - \varepsilon G &\sim 2T + f^*D - \varepsilon G \\ &\sim 2T + f^*D - p\varepsilon T \\ &\sim (2 - p\varepsilon)\, T + f^*D \end{aligned}$$

Since $\delta := 2 - p\varepsilon \in (0, 1)$, we have

$$-K_{P/Y} - \varepsilon G = \delta T + f^*D = \delta(T + f^*D) + (1 - \delta)f^*D.$$

Since $T + f^*D$ is f-ample and nef and D is ample, we see that $-K_{P/Y} - \Delta$ is ample. Note that the coefficient of $G|_{P_{\overline{\eta}}}$ is p, so we see that $(P_{\overline{\eta}}, \varepsilon G|_{P_{\overline{\eta}}})$ is not F-pure.

From here on, we assume that $p = 2$ or 3. Put $l := 5 - p$.

Example 2.4 ($p = 2, 3$)**.** We construct an algebraic fiber space $f : X \to Y$ such that X is a canonical projective 3-fold having a crepant resolution, that Y is a smooth projective curve and that $-K_{X/Y}$ is ample.

Let $C = Y$, D, $\mathcal{E}$ be as in the before Example 2.3. Set $\mathcal{G} := \mathcal{E} \oplus \mathcal{O}_Y(-D)$ and $P := \mathbb{P}(\mathcal{G})$. Let $h : P \to Y$ be the projection. Let F_1 be the prime divisor on P corresponding to the direct summand $\mathcal{O}_Y(-D) \subset G$. Let G' be the prime divisor on $\mathbb{P}(F^*\mathcal{G})$ corresponding to subbundle $\mathcal{O}_Y \subset F^*\mathcal{G}$, and let $G := F^*_{P/Y}G'$. Note that F_1 and G are intersect, which is different from Example 2.3. Put $T := c_1(\mathcal{O}_P(1))$. Since we have

$$F_1 + G \sim (p+1)T + h^*D \sim l(rT + h^*D'),$$

where $r := \frac{p+1}{l}$ and $D' := \frac{1}{l}D$. Note that D' is integral. From this, we can take the l-cyclic cover $\pi : X \to P$ brunched over $F_1 + G$. Then,

$$\begin{aligned} -K_{X/Y} &\sim \pi^*(-K_{P/Y} - (l-1)(rT + D')) \\ &\sim \pi^*(3T + 2f^*D - (l-1)(rT + D')) \\ &\sim \pi^*\left((3 - (l-1)r)T + \frac{l+1}{l}f^*D\right) \\ &\sim \pi^*\left(T + f^*(D + D')\right). \end{aligned}$$

Since $\mathcal{E}(D + D')$ is an ample vector bundle, we see that $-K_{X/Y}$ is ample.

Next, we verify that X has a crepant resolution. Let Σ be the section of $h : P \to Y$ corresponding to $\mathcal{G} \twoheadrightarrow \mathcal{O}_Y(-D)$ and put $\Sigma_X := (\pi^{-1}(\Sigma))_{\mathrm{red}}$. Then one can check that

$$\mathrm{Sing}(X)_{\mathrm{red}} = (\pi^{-1}(F_1 \cap G))_{\mathrm{red}} \sqcup \Sigma_X.$$

Since $(\pi^*F_1)_{\mathrm{red}}$ and $(\pi^*G)_{\mathrm{red}}$ is SNC along over $F_1 \cap G$, so the local equation of X is

$$u^l = vw,$$

so X is étale locally

$$(A_{l-1} \text{ surface singularities}) \times (\text{smooth curve})$$

over $F_1 \cap G$, which has a crepant resolution.

As for around Σ_X, the local equation of X is

$$u^l = x^p + ty^p.$$

When $(p, l) = (2, 3)$, we obtain

$$u^3 = x^2 + ty^2.$$

The blow-up along the ideal (x, y, u) is smooth and the discrepancy is 0. Thus, when $(p, l) = (2, 3)$, the crepant resolution $\tilde{X} \to X$ can be obtained by the blow-up along $\mathrm{Sing}(X)_{\mathrm{red}}$.

When $(p, l) = (3, 2)$, the local equation of X around Σ_X is

$$u^2 = x^3 + ty^3.$$

The y-chart of the same blow-up as above is

$$w^2 = y(v^3 + t)$$

with discrepancy 0. Taking $s = v^3 + t$, we get $w^2 = ys$, which is again $A_1\times$(smooth curve). Hence it has a crepant resolution. Note that, unlikely to the case $(p, l) = (2, 3)$, we need blowing-up 2 times in this case.

Example 2.5 ($p = 2, 3$)**.** We construct an algebraic fiber space $f : X \to Y$ such that X is an lc 3-fold, that Y is a smooth projective curve and that $-K_{X/Y}$ is ample. Let $C = Y$, D, $\mathcal{E}$, $\mathcal{G}$, $h : P \to Y$, T, F_1, G, r and D' be as in Example 2.4. Let F_2 be the prime divisor on P corresponding to the subbundle $\mathcal{O}_Y(-D) \subset \mathcal{E} \subset \mathcal{G}$ of $\mathcal{G}$. Since we have

$$F_2 + G \sim (p+1)T + h^*D \sim l(rT + h^*D')$$

we can take the l-cyclic cover $\pi : X \to P$ brunched over $F_2 + G$. Then, by the same way as that in Example 2.4, we see that $-K_{X/Y}$ is ample.

Next, we check that X is lc. Let Σ and Σ_X be as in Example 2.4. Then $\Sigma = (F_2 \cap G)_{\mathrm{red}}$ and $F_2 + G$ is smooth outside of Σ, so we have

$$\mathrm{Sing}(X)_{\mathrm{red}} = \Sigma_X.$$

Let $\mu : \tilde{P} \to P$ be the blow-up of P along Σ, and let E be its exceptional divisor. Put $B := F_2 + G$ and let $\tilde{B}$ be the strict transform of B. Then

$$K_{\tilde{P}} = \mu^* K_P + E \quad \text{and} \quad \mu^* B = \tilde{B} + (p+1)E,$$

and so

$$K_{\tilde{P}} + \frac{l-1}{l}\tilde{B} = \mu^*\left(K_P + \frac{l-1}{l}B\right) + \left(1 - \frac{l-1}{l}(p+1)\right)E$$

Let $\tilde{X}$ be the normalization of $X \times_P \tilde{P}$. We use the following notation:

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{\beta} & X \\ {\scriptstyle\tilde{\pi}}\downarrow & & \downarrow{\scriptstyle\pi} \\ \tilde{P} & \xrightarrow[\mu]{} & P. \end{array}$$

Let R be (the sum of) the reduced exceptional divisor of β. By the local computation, we see that

- $\tilde{\pi} : \tilde{X} \to \tilde{P}$ is an l-cyclic cover brunched over $\tilde{B}$,
- $\tilde{\pi}^* E = R$, and
- $\tilde{\pi}|R : R \to E$ is an l-cyclic cover brunched over $\tilde{B}|_E$. In particular, R is smooth, since $\tilde{B}|_E$ is smooth.

Therefore, by pulling-back the above equation, we obtain

$$K_{\tilde{X}} = \beta^* K_X + \left(\frac{p+1}{l} - p\right) R.$$

Since $\{p, l\} = \{2, 3\}$, we get

$$K_{\tilde{X}} + R = \beta^* K_X.$$

Since $\beta : \tilde{X} \to X$ is a log resolution, we see that X is lc and $\beta(R) = \Sigma_X$ is the unique lc center. By the construction, Σ_X is a section of f, so the proof completes.

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